POKHARA UNIVERSITY
FACULTY OF SCINCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2024		
Level	B.E.	F M	100
Program	BE Computer	PM	45
Year/ Part	IV/II	Time	3 Hrs

Subject: Digital Signal Analysis and Processing

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.

1.	**a)**	Define a system as Linear, Time Invariant, Causal and Stable. Check the given system for linearity, time invariance, causality and stability: y[n] = 5x[n] -2.	**7**
	b)	Consider the analog signal: $x_a(t)$ = 3cos (2000πt) + 5 sin (6000πt) + 10 cos (12000πt). i. What is Nyquist rate for this signal xa(t)? ii. Suppose that the signal is sampled at the rate of Fs=5000 samples per second, what is the discrete-time signal obtained after sampling? iii. What is the analog signal we can reconstruct from the samples? iv. What is the aliasing effect?	**8**
2.	**a)**	Verify the commutative property of convolution where input signal, x[n] = {1, 1, 1, 1} and response of system, h[n] = {1, 1, 1, 1}.	**8**
	b)	What are the two major characteristics of ROC? Determine the inverse z-transform of: $H(z) = \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}}$	**7**
3.	**a)**	Why do we need DFT even though we have DTFT? Explain DFT as a linear transformation.	**8**
	b)	Find the circular convolution of the sequence: $x(n) = \{0,1,2,3\}$ $h(n) = \{2\ 1,1,2\}$	**7**
4.	**a)**	Determine the 4-point DFT of the following sequence using DIT FFT radix-2 algorithm: $x(n) = u(n) - u(n-4)$	**8**
	b)	Draw the lattice structure from the given FIR filter's system function: $H(z) = 1 + \frac{13}{12}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$	**7**
5.	**a)**	Differentiate FIR with IIR system.	**5**
	b)	What do you mean by the term window in digital filter design? "Kaiser window is better than Hamming window." Justify this statement	**5**
	c)	Explain about Remez-Exchange algorithm for optimal digital filter design?	**5**

6.	**a)** The transfer function of an analog lowpass filter is $H_a(s) = \frac{1}{s+1}$, and its bandwidth is 1 rad/sec. Design the digital IIR filter using Bilinear transformation method whose cut-off frequency is 0.2π and sampling interval is 0.0167 sec, also consider frequency warping effect. **b)** Find the order and cut-off frequency of a digital filter by using impulse invariance method with the following specifications: $0.89 \leq \left\|H\left(e^{jw}\right)\right\| \leq 1$, for $0 \leq w \leq 0.4\pi$ $\left\|H\left(e^{jw}\right)\right\| \leq 0.18$, for $0.4\pi \leq w \leq \pi$ Also, plot its poles.	**7** **8**
7.	**Write short notes on:** *(Any two)* **a)** Advantages of digital over analog processor **b)** Discrete time Fourier series vs discrete time Fourier transform. **c)** Gibbs Phenomena in FIR filter design. **d)** Properties of Elliptic filter.	**2×5**

****** Best of Luck ******